[Dec-25]

GITAM (Deemed to be University)
[CSCI6131]
MCA Degree Examination

I Semester (Sessional –II)

OPERATING SYSTEMS: CONCEPTS AND DESIGN

(Effective from 2025–26 onwards)

Time: 2 Hour **Max.Marks: 45**

Instruction: All parts of the sections must be answered in one place only

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Section-A

Answer all the questions **(3×2 = 6M)**

1. List the conditions to be satisfied for a deadlock situation.
2. Give the contents of page table.
3. List the type of information maintained in an Access Control List (ACL).

Section-B

Answer the following **(3×5 = 15M)**

4. Identify and describe the key requirements that ensure a correct solution to the critical section problem.

OR

5. How are deadlocks described using Resource Allocation Graph? Illustrate with an example.

6. Differentiate between Physical and Logical Address space.

OR

7. Describe First fit, Best fit, and Worst fit techniques of memory allocation.

8. Analyze how different file access methods handle data organization and retrieval in an operating system.

OR

9. Describe in detail FCFS and SSTF Disk Scheduling concept.

Section-C

Answer the following **(3×8 = 24M)**

10. What is Readers Writers Problem? Analyze the role of the ‘mutex’ and ‘wrt’ semaphores in a typical Readers–Writers solution.

OR

11. How can a system recover from a deadlock? Describe the two available options.

12. Explain how memory protection and memory allocation are achieved in the Contiguous Memory Allocation technique.

OR

13. Explain how Effective Access Time (EAT) is calculated in a Demand Paged Memory Management system. Describe the sequence of steps involved in determining the page fault service time.

14. Describe the common schemes for defining the logical structure of a directory.

OR

15. Explain the physical structure and organization of a Magnetic Disk with a neat diagram.

[RB/1S/12/208/M]